Def. A Complete Normed Space X is Called a Banach Space.

Thm. Let X be a Normed Space. Then, the followings are equivalent:

1). X is Complete.

2). If a sequence $\{x_n\}$ in X satisfies $\sum_{n=1}^{\infty} \|x_n\| < \infty$, then $\sum_{n=1}^{\infty} x_n$ Converges.

Proof.